

Year 10
Mini Test 6 Pathway 1
Statistics



Baldivis
Secondary College

Resources allowed: Calculator and 1 x A4 page one side

Name: _____ Class: _____ Time: 55 minutes Total Marks: 70 69

Show working and answers on this sheet. Show working in sufficient detail to support your answers.

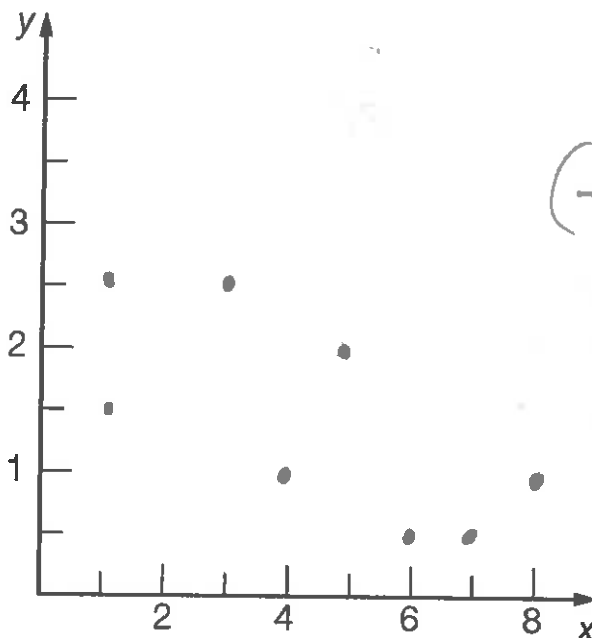
Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1

(2,2 – 4 Marks)

a) Plot the following points onto the graph.

(1, 1.5), (1, 2.5), (3, 2.5), (4, 1), (5, 2), (6, 0.5), (7, 0.5), (8, 1)



(-½ each error)

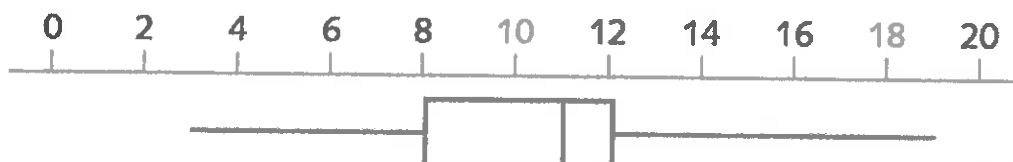
b) Describe the relationship between x and y.

Weak negative correlation.

Question 2

(1,1,1,1 – 4 Marks)

In a market survey, 200 people were asked how many hours of television they watched in the previous week. The results are presented in the boxplot below.



a) What is the maximum number of hours anyone watched television? *19 ✓*

b) How many people watched more than 8 hours of television? *150 ✓*

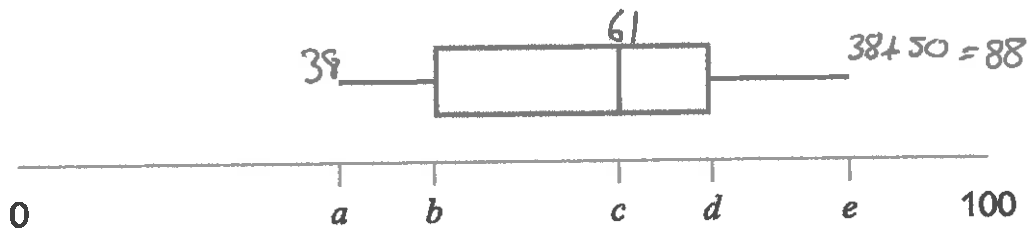
c) What is the interquartile range? *12 - 8 = 4 ✓*

d) How many people watched between 8 hours and 11 hours of television? *50 ✓*

Question 3

(3,2 – 5 Marks)

The boxplot shows the distribution of test scores in a class (Class A) of 20 students.



The lowest score in the class was 38, the range of the scores was 50 and the median was 61.

a) Write down the values of

$$a = 38 \checkmark$$

$$c = 61 \checkmark$$

$$e = 88 \checkmark$$

b) When all the test scores were added up the total was 1240. What was the mean of the test scores?

$$\begin{aligned} \text{Mean} &= \frac{1240}{20} \checkmark \\ &= 62 \checkmark \end{aligned}$$

Question 4

(1, 1, 2, – 4 Marks)

The stem-and-leaf plot shows the distribution of test 4 47 scores in Class 10.7 for the same test.

a) Assuming all students sat for the test, write down the number of students in Class 10.7.

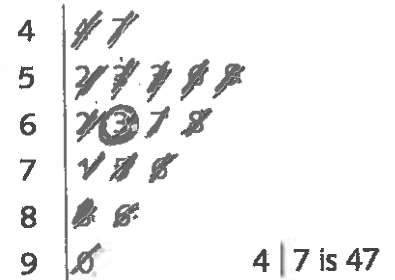
$$17 \checkmark$$

b) Find the median of the scores for Class 10.7.

$$63 \checkmark$$

c) Find the Mean Score for the Class

$$\frac{1105}{17} = 65 \checkmark$$



Question 5

(4 Marks)

The daily maximum temperatures recorded for a two-week period are shown in the table below.

Week 1	18.2	22.5	17.6	24.6	15.4	19.5	16.3
Week 2	25.6	28.2	29.0	28.5	33.2	28.8	30.4

- a) Find the mean and standard deviation of the temperatures for both weeks. Give your answers correct to one decimal place.

Wk 1: Mean = 19.2 ✓ SD = 3.1 ✓

Wk 2: Mean = 29.1 ✓ SD = 2.1/2.3 ✓

Question 6

(1, 1, 2, 2 -6 Marks)

The Cats and The Mice basketball teams compare their number of points per match for a season. The data is presented in this back-to-back stem-and-leaf plot.

The Cats Leaf	Stem	The Mice Leaf
	0	9
2	1	9
8 3	2	0 4 8 9
7 4	3	2 4 7 8 9
9 7 4 1 0	5	2 8
7 6 2	6	0

State which team has:

- a) the higher range

Cats ✓

- b) the higher median

Cats ✓

- c) the higher mean

Cats ✓

- d) the higher standard deviation

Cats ✓

Question 7

(3,2,2 -7 Marks)

The number of late passes issued daily by a school over a four week period were as follows:

8, 7, 7, 7, 9, 6, 5, 4, 14, 4, 6, 8, 10, 6, 5, 4, 8, 7, 9, 9

a) Determine the five number summary for this data.

4, 4, 4, 5, 5, 6, 6, 7, 7, 7, 8, 8, 8, 9, 9, 9, 10, 14

Min = 4

Max = 14

$Q_1 = 5.5$

$Q_2 = 7$

$Q_3 = 8.5$

(-1 each error)

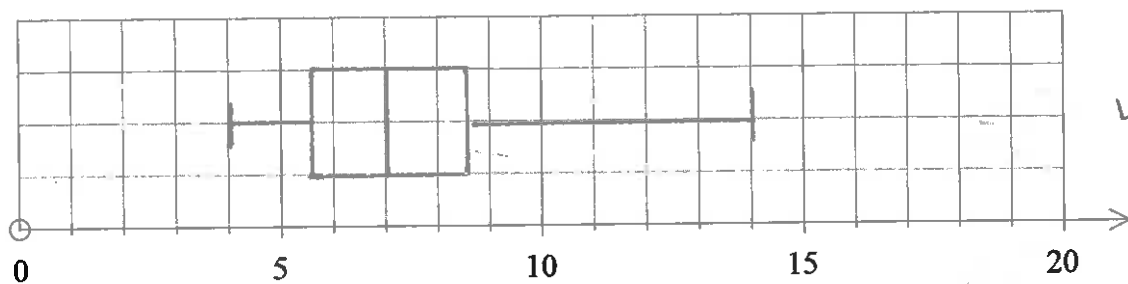
✓✓✓

b) Is the value of 14 in the above list an **outlier** for the data set? Justify your answer.

Yes ✓

Justification ✓

c) Use the five number summary to construct a box plot on the scale below that reflects your response in (b).



(-1/2 each error)

Question 8

(3 marks)

Two classes are marked on the same tests.

Class P has a mean of 65% and standard deviation of 5.

Class Q has a mean of 50% and standard deviation of 15.

- a) Write a statement comparing the scores of the two classes, that is who had the better scores?

P - higher average scores

Q - low average scores

- b) Write a statement comparing the spread of the scores for the two classes.

P - most scores are close to mean / tightly packed

Q - very spread out from the mean / spread out

- c) Which class would you expect to have the highest score? Explain.

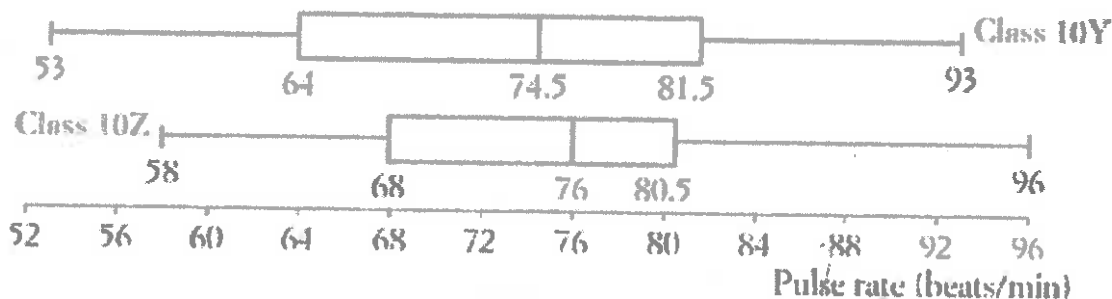
Q

$50 + 3 \times 15 > 65 + 3 \times 5$, Explanation

Question 9

(1,1,2,2,2 – 8 Marks)

Two PE classes of students have their pulse rates (in beats per minute) measured. The results are shown in these box plots.



- a) What was the lowest pulse rate across both classes?

53

- b) Find the interquartile range for class 10Z.

$$80.5 - 68 = 12.5$$

- c) If there are 24 students in class 10Y, how many had a pulse rate between 74.5 and 81.5?

$$24 \div 4 = 6$$

- d) Which class had the higher pulse rate on average? How does the box plot show this?

10Z

Highest Median

- e) Which class had more spread of pulse rates? How does the box plot show this?

Class 10Y

Biggest Range

Question 10

³
²
(5, 4, 4 - 9 Marks)

The data below gives the average monthly minimum daily temperatures of two Australian cities. The months are in order: January to December.

City X: 11.2, 11.8, 13.2, 9.3, 7.4, 5.8, 4.3, 5.9, 6.9, 19.8, 18.3, 12.4

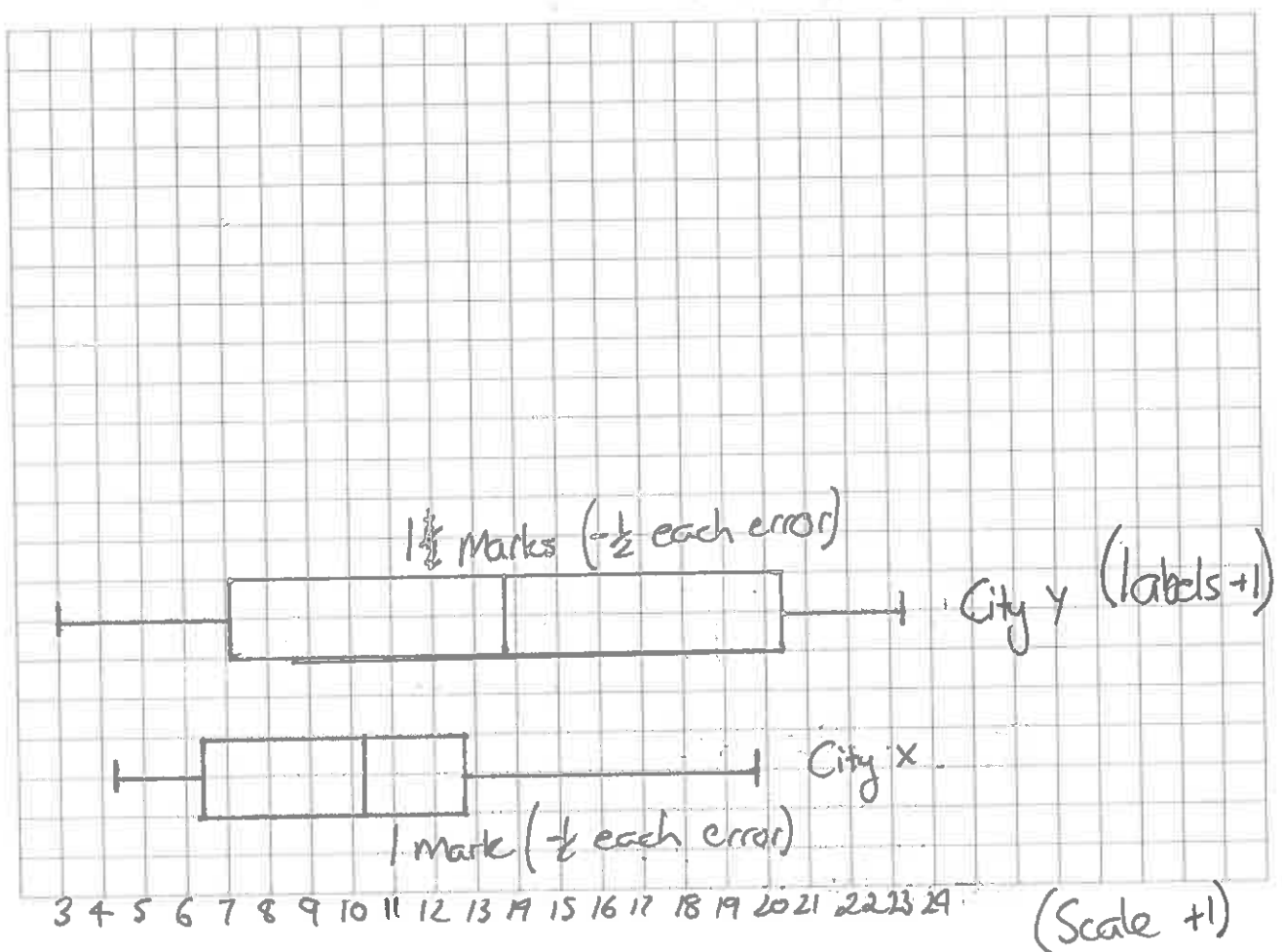
City Y: 23.2, 22.6, 18.4, 14.5, 7.2, 4.1, 3, 6.9, 8.7, 13.1, 18.3, 22.4

(a) Find the five-number summary for each city.

City X
Min = 4.3 $Q_2 = 10.25$
Max = 19.8 $Q_3 = 12.8$
 $Q_1 = 6.4$

CITY Y
Min = 3 $Q_2 = 13.8$ ($\frac{1}{2}$ each)
Max = 23.2 $Q_3 = 20.4$
 $Q_1 = 7.65$

(b) Draw a parallel box plot for the two cities.



(c) Write a couple of sentences comparing and contrasting the two cities.

Spread / Range
Medians
Min, Max
Skew

✓✓~~✗~~

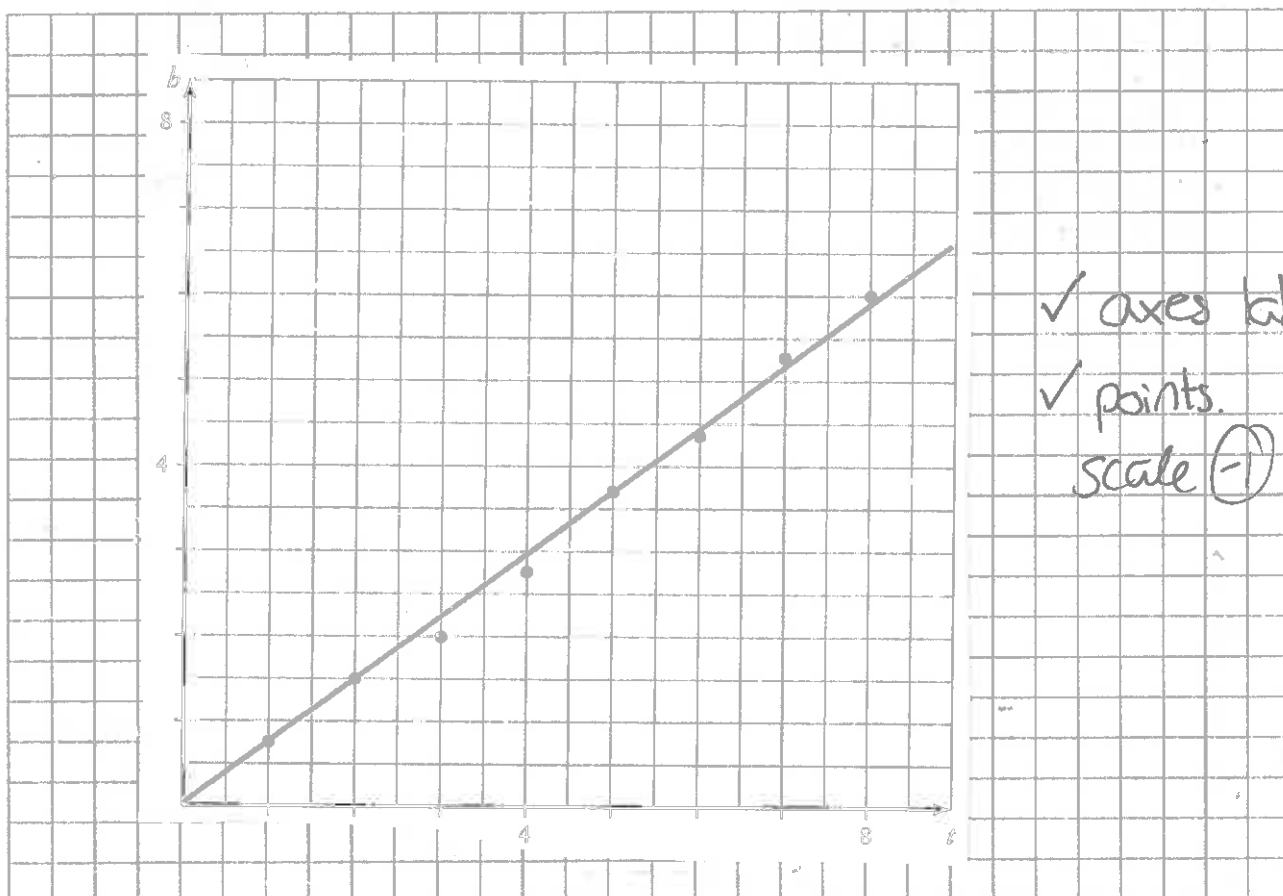
Question 11

(2,1,1,1,1 - 6 Marks)

The height of a seedling after the first shoot appears is recorded in the table below.

Day	1	2	3	4	5	6	7	8
Height (cm)	0.7	1.5	2.0	2.8	3.6	4.3	5.1	6.0

a) Plot the points in a scatter plot in the grid below.



b) Describe the relationship between the day and the height.

Strong positive ✓

c) Draw in the line of best fit.

✓

d) Use the graph to find the height after 2.5 days. What is this prediction called?

1.8cm^{1/2}, interpolation ✓

e) Use your answer to d to predict the height be after 10 days. What is this prediction called?

Question 12 (4, 4, 2 - 10 Marks)

A group of 16 people was surveyed to find the number of hours of television they watch in a week.

The raw data is listed: ~~6, 5, 11, 13, 24, 8, 1, 12~~ 1, 6, 14, 10, 9, 16, 8, 3

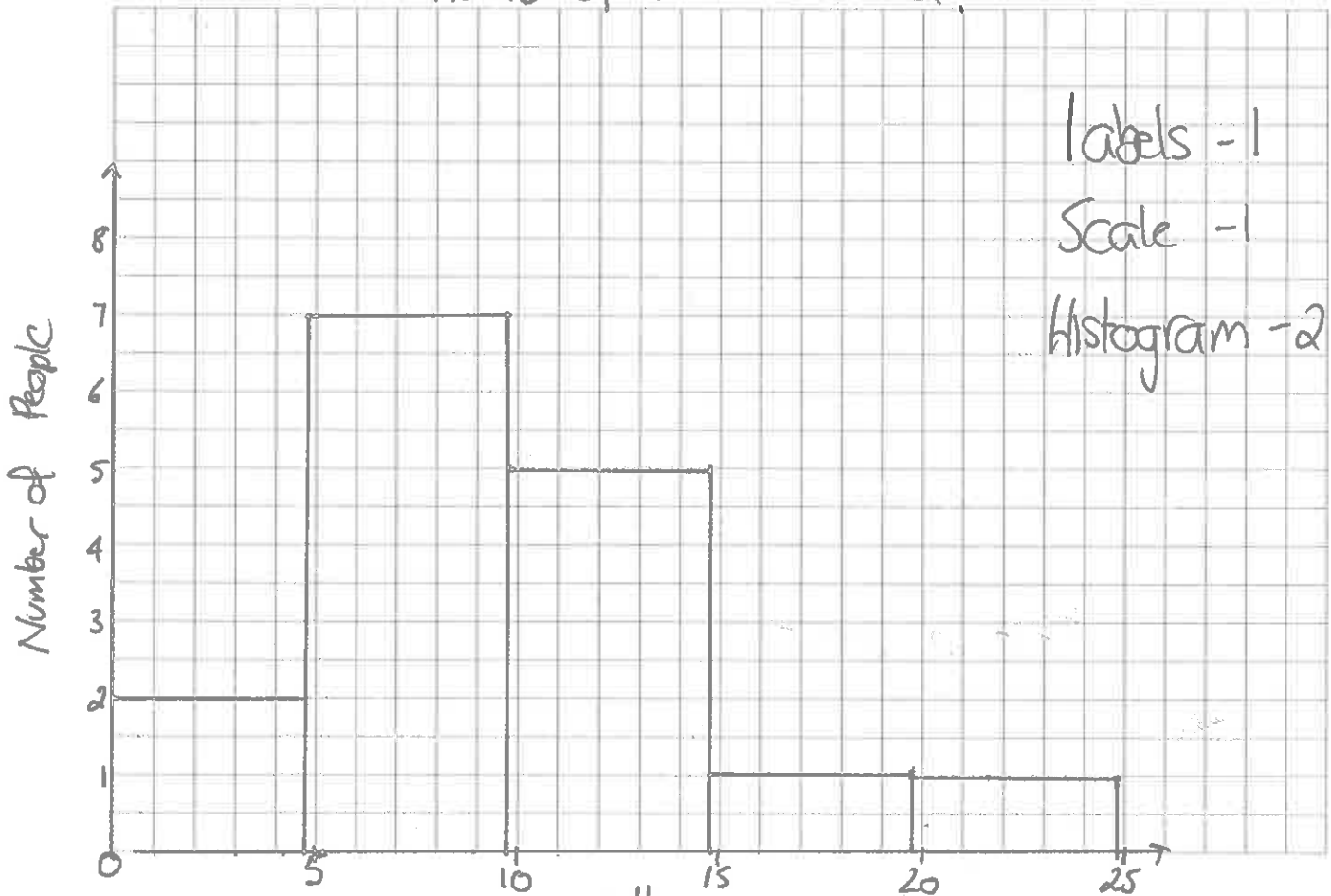
- a) Organise the data into a table with class intervals of 5.

Classes (hrs)	People
0 - 4	2
5 - 9	7
10 - 14	5
15 - 19	1
20 - 24	1

(-1/2 each error)

- b) Construct a histogram for the data.

Hours of TV Watched



- c) Would you describe the data as symmetrical, positively skewed or negatively skewed?

skewed right.

~End of Test~